Ronald J. **Tackett, Ph.D.**

Founding Head of School & Associate Professor of Physics
School of Foundational Studies · Kettering University

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EDUCATION

Ph.D., Condensed Matter Physics 2003–2008
Wayne State University, Detroit, MI

- Dissertation: *The Synthesis, Characterization, and Application of Multifunctional Magnetic Nanoparticles*
- Advisor: Dr. Gavin Lawes

B.S., Engineering Physics 2001–2003
Eastern Michigan University, Ypsilanti, MI

PROFESSIONAL EXPERIENCE

Founding Head (Dean-Equivalent), School of Foundational Studies 2025–pres.
Kettering University, Flint, MI

Adjunct (Volunteer) Associate Professor of Physics 2025–pres.
Wayne State University, Detroit, MI

Associate Professor of Physics 2019–pres.
Kettering University, Flint, MI

Assistant Professor of Physics 2014–2019
Kettering University, Flint, MI

Assistant Professor of Physics 2010–2014
Arkansas Tech University, Russellville, AR

Postdoctoral Research Associate 2008–2010
University of Texas at El Paso, El Paso, TX

Graduate Research Assistant 2004–2008
Wayne State University, Detroit, MI

Graduate Teaching Assistant 2003–2004
Wayne State University, Detroit, MI

PUBLICATIONS

Published

B.R. Dahal, K. Schroeder, M.M. Allyn, **R.J. Tackett**, Y. Huh, and P. Kharel. "Near-room-temperature magnetocaloric properties of $La_{1-x}Sr_xMnO_3$ (x = 0.11, 0.17, and 0.19) nanoparticles." *Mat. Res. Express* **5**, 106103 (2018). doi:10.1088/2053-1591/aadabd

R.J. Tackett, J. Thakur, N. Mosher, E. Perkins-Harbin, R.E. Kumon, L. Wang, C. Rablau, and P.P. Vaishnava. "A method for measuring the Néel relaxation time in a frozen ferrofluid." *J. Appl. Phys.* **118(6)**, 064701 (2015). doi:10.1063/1.4928202

Y.C.N. Chen, C.Y. Hsieh, **R. Tackett**, P. Kokeny, R.K. Regmi, and G. Lawes. "Magnetic moment quantifications of small spherical objects in MRI." *Mag. Res. Imaging* **33(6)**, 829 (2015). doi:10.1016/j.mri.2014.11.003

C.E. Botez, A.H. Adair, and **R.J. Tackett**. "Evidence of superspin-glass behavior in $Zn_{0.5}Ni_{0.5}Fe_2O_4$ nanoparticles." *J. Phys. Cond. Mat.* **27(7)**, 076005 (2015). doi:10.1088/0953-8984/27/7/076005

S.S. Laha, **R.J. Tackett**, and G. Lawes. "Interactions in γ-Fe_2O_3 and Fe_3O_4 nanoparticle systems." *Physica B* **448**, 69 (2014).

C.E. Botez, **R.J. Tackett**, J.D. Hermosillo, J.Z. Zhang, Y.S. Zho, and L.P. Wang. "High-pressure synchrotron x-ray diffraction studies of superprotonic transitions in phosphate solid acids." *Solid State Ionics* **213**, 58 (2012). doi:10.1016/j.ssi.2011.08.015

R. Tackett, G. Lawes, R. Suryanarayanan, M. Apostu, and A. Revcolevschi. "The zero-field glassy ground state and field-induced ferromagnetic transition in $(La_{0.4}Pr_{0.6})_{1.2}Sr_{1.8}Mn_2O_7$." *J. Phys. Cond. Mat.* **23**, 156004 (2011). doi:10.1088/0953-8984/23/15/156004

K.K. Bharathi, **R.J. Tackett**, C.E. Botez, and C.V. Ramana. "Coexistence of spin glass behavior and long-range ferrimagnetic ordering in La- and Dy-doped Co ferrite." *J. Appl. Phys.* **109**, 07A510 (2011). doi:10.1063/1.3562201

C.E. Botez, A.W. Bhuiya, and **R.J. Tackett**. "Dynamic-susceptibility studies of the interplay between the Néel and Brown magnetic relaxation mechanisms." *App. Phys. A* **104**, 177 (2011). doi:10.1007/s00339-010-6098-x

C.E. Botez, D. Carbajal, K. Adiraju, and **R.J. Tackett**. "Intermediate-temperature polymorphic phase transition in KH_2PO_4: A synchrotron x-ray diffraction study." *J. Phys. Chem. Sol.* **71**, 1576 (2010). doi:10.1016/j.jpcs.2010.08.004

R.J. Tackett, J.G. Parsons, B.I. Machado, S. Gaytan, L.E. Murr, and C.E. Botez. "Evidence of low-temperature superparamagnetism in Mn_3O_4 nanoparticle ensembles." *Nanotechnology* **21**, 365703 (2010). doi:10.1088/0957-4484/21/36/365703

R.J. Tackett, A.W. Bhuiya, and C.E. Botez. "Dynamic susceptibility evidence of surface spin freezing in ultrafine $NiFe_2O_4$ nanoparticles." *Nanotechnology* **20**, 445705 (2009). doi:10.1088/0957-4484/20/44/445705

C.E. Botez, H. Martinez, **R.J. Tackett**, R.R. Chianelli, J. Zhang, and Y. Zhao. "High-temperature crystal structures and chemical modifications in RbH_2PO_4." *J. Phys. Cond. Mat.* **21**, 325401 (2009). doi:10.1088/0953-8984/21/32/325401

K. Senevirathne, **R. Tackett**, P.R. Kharel, G. Lawes, K. Somaskandan, and S.L. Brock. "Discrete, dispersible MnAs nanocrystals from solution methods: phase control on the nanoscale and magnetic consequences." *ACS Nano* **3**, 1129 (2009). doi:10.1021/nn900194f

R. Regmi, **R. Tackett**, and G. Lawes. "Suppression of low-temperature magnetic states in Mn_3O_4 nanoparticles." *J. Magn. Magn. Mater.* **321**, 2296 (2009). doi:10.1016/j.jmmm.2009.01.041

B.C. Melot, **R. Tackett**, J. O'Brien, A.L. Hector, G. Lawes, R. Seshadri, and A.P. Ramirez. "Large low-temperature specific heat in pyrochlore $Bi_2Ti_2O_7$." *Phys. Rev. B* **79**, 224111 (2009). doi:10.1103/PhysRevB.79.224111

S. Putatunda, A. Singer, **R. Tackett**, and G. Lawes. "Development of a high-strength, high-toughness ausferritic steel." *Metall. Mater. Trans. A* **513–514**, 329 (2009). doi:10.1016/j.msea.2009.02.013

Y. Shen, Y.C.N. Cheng, G. Lawes, J. Neelavalli, C. Sudakar, **R. Tackett**, H.P. Ramanth, and E.M. Haacke. "Quantifying magnetic nanoparticles in non-steady flow by MRI." *Magn. Reson. Mater. Phys.* **21**, 345 (2008). doi:10.1007/s10334-008-0140-4

C. Rablau, P.P. Vaishnava, C. Sudakar, **R. Tackett**, G. Lawes, and R. Naik. "Magnetic field induced optical

anisotropy in ferrofluids: A time dependent light scattering investigation.” *Phys. Rev. E* **78**, 051502 (2008). doi:10.1103/PhysRevE.78.051502

R. Tackett, C. Sudakar, R. Naik, G. Lawes, C. Rablau, and P.P. Vaishnava. “Magnetic and optical response of tuning the magnetocrystalline anisotropy in Fe_3O_4 ferrofluids by Co doping.” *J. Magn. Magn. Mater.* **320**, 2755 (2008). doi:10.1016/j.jmmm.2008.06.006

S. Singh, R. Suryanarayanan, **R. Tackett**, G. Lawes, A.K. Sood, P. Berthet, and A. Revcolevschi. “Ordered spin-ice state in the geometrically frustrated metallic ferromagnet $Sm_2Mo_2O_7$.” *Phys. Rev. B* **77**, 020406 (2008). doi:10.1103/PhysRevB.77.020406 doi:10.1103/PhysRevB.77.020406

P.P. Vaishnava, **R. Tackett**, A. Dixit, C. Sudakar, R. Naik, and G. Lawes. “Magnetic relaxation and dissipative heating in ferrofluids.” *J. Appl. Phys.* **102**, 063914 (2007). doi:10.1063/1.2784080

R. Tackett, G. Lawes, B.C. Melot, M. Grossman, E.S. Toberer, and G. Lawes. “Magnetodielectric coupling in Mn_3O_4.” *Phys. Rev. B* **76**, 024409 (2007). doi:10.1103/PhysRevB.76.024409

J. Cao, R.C. Rai, S. Brown, J. Musfeldt, **R. Tackett**, G. Lawes, Y. Wang, X. Wei, M. Apostu, R. Suryanarayanan, and A. Revcolevschi. “Observation of 300 K high energy magnetodielectric response in the bilayer manganite $(La_{0.4}Pr_{0.6})_{1.2}Sr_{1.8}Mn_2O_7$.” *Appl. Phys. Lett.* **91**, 021913 (2007). doi:10.1063/1.2757120

S.K. Putatunda, S. Kesani, **R. Tackett**, and G. Lawes. “Development of austenite-free ADI (austempered ductile cast iron).” *Mater. Sci. Eng. A* **435–436**, 112 (2006). doi:10.1016/j.msea.2006.07.051

G. Lawes, **R. Tackett**, O. Masala, B. Adhikary, R. Naik, and R. Seshadri. “Positive and negative magnetocapacitance in magnetic nanoparticle systems.” *Appl. Phys. Lett.* **88**, 242903 (2006). doi:10.1063/1.2213194

G. Tsoi, U. Seneratne, **R. Tackett**, E. Buc, R. Naik, P. Vaishnava, V. Naik, and L. Wenger. “Memory effects and magnetic interactions in γ-Fe_2O_3 nanoparticle system.” *J. Appl. Phys.* **97(10)**, 10J507 (2005). doi:10.1063/1.1853898

G. Tsoi, L. Wenger, U. Seneratne, **R. Tackett**, E. Buc, R. Naik, P. Vaishnava, and V. Naik. “Memory effects in a superparamagnetic γ-Fe_2O_3 system.” *Phys. Rev. B* **72**, 014445 (2005).

AWARDS, FELLOWSHIPS, & GRANTS

Distinguished Service Award 2025
Kettering University

Strategic Investment Grant (Co-PI) 2024
Michigan Economic Development Corporation

Outstanding New Research Award 2018
Kettering University

Provost’s Research Matching Fund 2017
Kettering University

Faculty Research Fellowship 2016
Kettering University

Provost’s Matching Fund 2016
Kettering University

Provost’s Travel Fund 2016
Kettering University

Provost’s Travel Fund 2015
Kettering University

Major Research Instrumentation (1531402) 2015
National Science Foundation

Faculty Research Fellowship 2014
Kettering University

Undergraduate Research Award 2013
Arkansas Tech University

Junior Science and Humanities Symposium 2013
The Academy of Applied Science

Arkansas Space Grant Symposium 2012
NASA

Junior Science and Humanities Symposium 2012
The Academy of Applied Science

Arkansas Space Grant Symposium 2011
NASA

Undergraduate Research Award 2010
Arkansas Tech University

PRESENTATIONS

Invited Talks

2026. *In the Age of AI, What Are We Actually Educating Students For?* Second Gathering, Council of Liberal Arts Leaders at Technological Universities (CoLLT), Georgia Institute of Technology, Atlanta, GA.

2026. *The School of Foundational Studies at Kettering University: An Institutional Profile.* Pre-Planning Meeting, Council of Liberal Arts Leaders at Technological Universities (CoLLT), Georgia Institute of Technology, Atlanta, GA.

2024. *A Journey Through Innovation and Impact: How Semiconductors Revolutionized the World.* Longway Planetarium, Flint, MI.

2023. *Through the Looking Glass: A Discussion of the Art of Microscopy.* Longway Planetarium, Flint, MI.

2019. *Specialization and Optimization of Metal-Oxide Nanoparticles for Use in the Magnetic Hyperthermia Treatment of Cancer.* Fall Meeting, Midland Section, American Chemical Society, Saginaw Valley State University, Saginaw, MI.

2018. *Tuesday Teaching Talk: EndNote.* Center for Excellence in Teaching and Learning, Kettering University, Flint, MI.

2017. *Through the Magnetic Lens: A Discussion of Electron Microscopy.* Science on Tap, Tenacity Brewing Company, Flint, MI.

2016. *Treating Big Problems with Small Things: Applications of Magnetic Nanoparticles.* Gavin Lawes Memorial Symposium, Wayne State University, Detroit, MI.

2016. *Treating Big Problems with Small Things: Applications of Magnetic Nanoparticles.* SPS Guest Lecture Series, Lawrence Technological University, Southfield, MI.

2015. *Treating Big Problems with Small Things: Applications of Magnetic Nanoparticles.* Distinguished Faculty Series, Kettering University, Flint, MI.

2014. *Magnetic Fluid Hyperthermia Treatment of Cancer using Dextran-Coated Fe_3O_4 Nanoparticles.* Solid State Physics Seminar, Wayne State University, Detroit, MI.

2009. *It's a Small World After All: Characterization and Applications of Magnetic Nanoparticles.* The University of Texas at El Paso, El Paso, TX.

2009. *High Temperature Phase Transitions in CsH_2PO_4 and RbH_2PO_4.* New Mexico State University, Las Cruces, NM.

2006. *The Synthesis and Characterization of Magnetic Nanoparticles.* Université Paris-Sud, Orsay, France.

Contributed Presentations

J. Wylie and **R.J. Tackett**. 2019. Synthesis and Characterization of Gadolinium-Doped Magnetite. Fall Meeting, Eastern Great Lakes Section, APS, Kettering University, Flint, MI.

J. Wylie and **R.J. Tackett**. 2019. Synthesis and Characterization of Gadolinium-Doped Magnetite. Michigan Academy of Science, Arts, and Letters (MASAL), Alma College, Alma, MI.

M. Warteman and **R.J. Tackett**. 2019. Effects of Doping Concentration in the Sol-Gel Synthesis of Strontium-Doped Lanthanum Manganate. MASAL, Alma College, Alma, MI.

R.J. Tackett, M.M. Allyn, V.K. Garg, A.C. de Oliveira, and P.P. Vaishnava. 2017. A Comparison of Methods for the Determination of the Magnetocrystalline Anisotropy Constant. Spring Meeting, Ohio Section, APS, Eastern Michigan University, Ypsilanti, MI.

M.M. Allyn, P. Kharel, P.P. Vaishnava, and **R.J. Tackett**. 2017. The Investigation of Smart Magnetic Nanoparticles for Use in the Hyperthermia Treatment of Cancer. Spring Meeting, Ohio Section, APS, Eastern Michigan University, Ypsilanti, MI.

R.J. Tackett, M.M. Allyn, V.K. Garg, A.C. de Oliveira, E.A. Elhamid, and P.P. Vaishnava. 2016. A Comparison of Methods for the Determination of the Magnetocrystalline Anisotropy Constant. March Meeting, APS, Baltimore, MD.

M.M. Allyn, P. Kharel, P.P. Vaishnava, and **R.J. Tackett**. 2016. The Investigation of Smart Magnetic Nanoparticles for Use in the Hyperthermia Treatment of Cancer. March Meeting, APS, Baltimore, MD.

M.M. Allyn, P. Kharel, P.P. Vaishnava, and **R.J. Tackett**. 2016. The Investigation of Smart Magnetic Nanoparticles for Use in the Hyperthermia Treatment of Cancer. MASAL, Saginaw Valley State University, Saginaw, MI.

N. Mosher, E. Perkins-Harbin, B. Aho, L. Wang, R.E. Kumon, C. Rablau, P.P. Vaishnava, and **R.J. Tackett**. 2015. Determination of the Magnetocrystalline Anisotropy Constant from Frequency-Dependence of the Specific Absorption Rate in a Frozen Ferrofluid. March Meeting, APS, San Antonio, TX.

J.S. Pennington and **R.J. Tackett**. 2013. Variant Crystal Lattice Structure of γ-Fe_2O_3 Under Successive Loadings Into an Alginate Polymer Matrix. Arkansas Space Grant Consortium, Morrilton, AR; Arkansas Academy of Science, Little Rock, AR.

J.D. Counts and **R.J. Tackett**. 2013. Particle Size Dependence of Superparamagnetic Blocking in Magnetite (Fe_3O_4) Nanoparticles. Arkansas Space Grant Consortium, Morrilton, AR.

J.D. Counts and **R.J. Tackett**. 2012. Particle Size Dependence of Superparamagnetic Blocking in Fe_3O_4 Nanoparticles. Arkansas Academy of Science, Southern Arkansas University, Magnolia, AR; Senior Honors Symposium, Arkansas Tech University, Russellville, AR.

J.C. Smith and **R.J. Tackett**. 2011. Simulation of Powder X-Ray Diffraction Using a Polycrystalline Film of Spherical Colloids. Arkansas Academy of Science, University of Arkansas at Monticello, AR.

R.J. Tackett, H. Martinez, R.R. Chianelli, J. Zhang, Y. Zhao, and C.E. Botez. 2010. High-Temperature Crystal Structures and Chemical Modifications in RbH_2PO_4. March Meeting, APS, Portland, OR.

C. Rablau, P.P. Vaishnava, R. Naik, G. Lawes, **R. Tackett**, and C. Sudakar. 2008. Tuning of the Magnetocrystalline Anisotropy in $Co_xFe_{3-x}O_4$ Nanoparticles. March Meeting, APS, New Orleans, LA.

R. Tackett, C. Sudakar, R. Naik, G. Lawes, C. Rablau, and P.P. Vaishnava. 2008. March Meeting, APS, New Orleans, LA.

R. Tackett, E. Toberer, R. Seshadri, and G. Lawes. 2007. Magnetodielectric Coupling in Mn_3O_4. March Meeting, APS, Denver, CO.

R. Tackett, O. Masala, B. Adhikary, R. Naik, A.P. Ramirez, R. Seshadri, and G. Lawes. 2006. The Coupling of Magnetic and Dielectric Properties in Magnetic Nanoparticles. March Meeting, APS, Los Angeles, CA.

TEACHING EXPERIENCE

Kettering University

Introduction to Semiconductors SCI-100
Lead Instructor

Introduction to Material Science & Engineering EP-342
Lead Instructor

Solid State Physics EP-446
Lead Instructor

College Mathematics MATH-100
Lead Instructor

Newtonian Mechanics PHYS-114
Lead Instructor

Newtonian Mechanics Laboratory PHYS-115
Lead Instructor

Electricity & Magnetism PHYS-224
Lead Instructor

Electricity & Magnetism Laboratory PHYS-225
Lead Instructor

The Science of Sensors PHYS-304
Lead Instructor

Thermodynamics & Statistical Physics PHYS-452
Lead Instructor

Quantum Mechanics PHYS-462
Lead Instructor

Arkansas Tech University

Physical Science Laboratory PHSC-1021
Lead Instructor

Introduction to Physical Science PHSC-1013
Lead Instructor

Physics Laboratory I PHYS-2000
Lead Instructor

Physics Laboratory II PHYS-2010
Lead Instructor

Algebra-Based Physics I PHYS-2014
Lead Instructor

Algebra-Based Physics II PHYS-2024
Lead Instructor

Calculus-Based Physics I PHYS-2114
Lead Instructor

Calculus-Based Physics II PHYS-2124
Lead Instructor

Solid State Physics PHYS-3153
Lead Instructor

Advanced Physics Laboratory PHYS-4113
Lead Instructor

RESEARCH MENTORING

Bradley Henry 2021
Senior Research Thesis, Kettering University

Joshua Wiley 2020
Senior Research Thesis, Kettering University

Przemyslaw Piotrowski 2019
Senior Research Thesis, Kettering University

Megan Allyn 2018
Senior Research Thesis, Kettering University

Joy Counts 2013
Undergraduate Research Assistant, Arkansas Tech University

Joshua Pennington 2012
Undergraduate Research Assistant, Arkansas Tech University

SERVICE

Service to the Field

Treasurer, Eastern Great Lakes Section 2026–pres.
American Physical Society

Past Chair, Eastern Great Lakes Section 2026–pres.
American Physical Society

Founding Member 2026–pres.
Council of Liberal Arts Leaders at Technological Universities

Chair, Eastern Great Lakes Section 2025–2026
American Physical Society

Advocacy Champion 2025–pres.
American Physical Society

Chair-Elect, Eastern Great Lakes Section 2024–2025
American Physical Society

Vice Chair, Eastern Great Lakes Section 2023–2024
American Physical Society

Review Panelist (P232162) 2023
National Science Foundation

Reviewer 2021–pres.
AIP Advances

Reviewer 2020–pres.
Journal of Alloys and Compounds (Elsevier)

Reviewer 2018–pres.
American Journal of Physics (AAPT)

Reviewer 2017–2020
NSF Graduate Research Fellowship Program

Reviewer 2017–pres.
The Physics Teacher (AAPT)

Reviewer 2016–2017
National Defense Science and Engineering Graduate Fellowship

Service to the University

Reviewer, President's Medal Nominations 2025
Kettering University

Founding Chair, Instructional Technology Advisory Committee 2024–2025
Kettering University

Moderator, Faculty Senate 2024–2025
Kettering University

Moderator-Elect, Faculty Senate 2023–2024
Kettering University

Member, Employee Satisfaction and Fulfillment Working Group 2022–2023
Kettering University

Senator (Natural Sciences), Faculty Senate 2020–2023
Kettering University

Chair, Faculty Senate Thesis Committee 2020–2023
Kettering University

Secretary, Faculty Senate 2019–2020
Kettering University

Member, Advisory Board for Excellence in Teaching & Learning 2019–2025
Kettering University

Member, University Curriculum Committee 2016–2018
Kettering University

Service to the Community

Commencement Speaker 2026
Livonia Stevenson High School

Volunteer (Pit/Props) 2025–pres.
Grand Blanc High School Marching Band

Facilitator, Quantum Science Workshop 2025
Genesee Career Institute

Secretary, Brendel Elementary PTO 2020–2023
Grand Blanc, MI

Member, Zero-Mil Increase Bond Committee 2020
Grand Blanc Community Schools

Presenter, Career Day 2019, 2020
Brendel Elementary School, Grand Blanc, MI

Professional Society Membership

American Physical Society (APS) 2008–pres.
Member

American Association of Physics Teachers (AAPT) 2015–pres.
Member

American Association for the Advancement of Science (AAAS) 2019–pres.
Member

Materials Research Society (MRS) 2019–2021
Member

Advanced Laboratory Physics Association (ALPhA) 2019–pres.
Member